Def. If $s: X \to [0,\infty)$ is a measurable simple function with representation

$$s = \sum_{i=1}^{n} a_i \cdot \chi_{A_i} \quad \text{where} \quad \text{range}(s) = \{a_1, \cdots, a_n\}.$$

For each $E \in \mathcal{M}$, define

$$\int_E s \, d\mu \overset{\text{def}}{=} \sum_{i=1}^{n} a_i \cdot \mu(A_i \cap E).$$

For a measurable function $f: X \to [0,\infty]$, and $E \in \mathcal{M}$, define

$$\int_E f \, d\mu \overset{\text{def}}{=} \sup \left\{ \int_E s \, d\mu \;\middle|\; s \text{ is a simple function s.t } 0 \le s \le f \right\}.$$

This integral is called a Lebesgue integral of f over E, with respect to the measure μ.

Prop. Let $(X, \mathcal{M}, \mu)$ be a measure space, let $f, g: X \to [0,\infty]$ be measurable.

1). If $f \le g$, then $\int_E f \, d\mu \le \int_E g \, d\mu$.

2) If $A \subset B$, then $\int_A f \, d\mu \le \int_B f \, d\mu$

3). If $c \in [0,\infty)$, then $\int_E c \cdot f \, d\mu = c \cdot \int_E f \, d\mu$.

4). If $f|_E \equiv 0$, then $\int_E f \, d\mu = 0$.

5). If $\mu(E) = 0$, then $\int_E f \, d\mu = 0$.

6). $\int_E f \, d\mu = \int_X \chi_E \cdot f \, d\mu$.

Prop. Let s be nonnegative measurable simple function on X.
For each $E \in \mathcal{M}$, define $\varphi(E) \overset{\text{def}}{=} \int_E s\, d\mu$.

Then, the map $\varphi: \mathcal{M} \to [0,\infty]$ is a measure on $(X, \mathcal{M})$ ⌋

Proof. Let $E_n \in \mathcal{M}$ $(n \in \mathbb{N})$ be pairwise disjoints.
Then, for $\mathrm{Im}(s) = \{a_1, \cdots, a_n\}$ and $s^{-1}[\{a_i\}] = A_i$,

$$\varphi\left(\bigcup_{n\in\mathbb{N}} E_n\right) \overset{\text{def}}{=} \sum_{i=1}^{n} a_i \cdot \mu\left(A_i \cap \left(\bigcup_{n\in\mathbb{N}} E_n\right)\right) \qquad A_i \cap \bigcup E_n = \bigcup A_i \cap E_n$$

$$= \sum_{i=1}^{n} a_i \cdot \sum_{r=1}^{\infty} \mu(A_i \cap E_r) \qquad \swarrow \mu \text{ is countably additive}$$

$$= \sum_{r=1}^{\infty} \sum_{i=1}^{n} a_i \cdot \mu(A_i \cap E_r) \qquad \swarrow \text{lemma}$$

$$= \sum_{r=1}^{\infty} \varphi(E_r)$$

And, $\varphi(\emptyset) = 0$, so that φ satisfies definitions of measure. ☑

Lemma. Let $\{s_n\}, \{t_n\}$ be increasing sequences in $[0,\infty]$. Then,

$$\lim_{n\to\infty}(s_n + t_n) = \left(\lim_{n\to\infty} s_n\right) + \left(\lim_{n\to\infty} t_n\right)$$

Proof. Since both are increasing,

$$\lim_{n\to\infty} s_n = \sup_{n\in\mathbb{N}} s_n \quad \text{and} \quad \lim_{n\to\infty} t_n = \sup_{n\in\mathbb{N}} t_n \quad \text{and} \quad \lim_{n\to\infty}(s_n+t_n) = \sup_{n\in\mathbb{N}}(s_n+t_n)$$

And, by the monotonicity, $\geq$ always, $\leq$ by mono.

$$\sup_{n\in\mathbb{N}} s_n + \sup_{n\in\mathbb{N}} t_n = \sup_{n,m\in\mathbb{N}}(s_n + t_m) = \sup_{n\in\mathbb{N}}(s_n + t_n), \text{ so proved.}$$

(For any $N, M \in \mathbb{N}$, take $K = \max\{N, M\}$. Then, by the monotonicity

$$s_N + t_M \leq s_K + t_K \leq \sup_{n\in\mathbb{N}}(s_n + t_n),$$

so $\sup_{n,m}(s_n + t_m) \leq \sup_n (s_n + t_n)$.

Debric.
(Given $\varepsilon > 0$, $\sup_{n,m}(s_n+t_m) - \varepsilon < s_N + t_M \leq \sup_{n,m}(s_n+t_m)$ for some $N, M \in \mathbb{N}$.
Take $K = \max\{N, M\}$. Then, $s_N + t_M \leq s_K + t_K \leq \sup_n (s_n + t_n)$. That is,

$$\sup_{n,m\in\mathbb{N}}(s_n + t_m) - \varepsilon \leq \sup_{n\in\mathbb{N}}(s_n + t_n).$$

Since $\varepsilon > 0$ was arbitrary, we obtain the inequality.) ☑
이거 빼도돼. $s_n = n$, $t_n = 0$ 이라고 $\sup s_n + t_m = \infty$

[Lebesgue's Monotone Convergence Theorem.
Let $\{f_n : X \to [0,\infty]\}$ be a sequence of measurable functions on $(X, \mathcal{M}, \mu)$.
Suppose that

$0 \le f_1 \le f_2 \le \cdots \le \infty$ and $f_n \to f$ pointwise on X.

Then, f is measurable with

$$\lim_{n\to\infty} \int_X f_n \, d\mu = \int_X \lim_{n\to\infty} f_n \, d\mu = \int_X f \, d\mu$$]

Proof. Since $f_n \le f_{n+1}$, so $\int f_n \le \int f_{n+1}$ and so that there exists $\alpha \in [0,\infty]$

$$\int_X f_n \, d\mu \to \alpha \text{ as } n \to \infty.$$

Meanwhile, $f = \lim_{n\to\infty} f_n = \sup_{n\in\mathbb{N}} f_n$, so f is measurable, and

$f_n \le f$ implies $\int_X f_n \, d\mu \le \int_X f \, d\mu$, thus $\alpha \le \int_X f \, d\mu$.

Let $s : X \to [0,\infty]$ be any simple measurable function such that $0 \le s \le f$, let $c \in (0,1)$ be a constant. For each $n \in \mathbb{N}$, define

$$E_n \overset{\text{def}}{=} \{x \in X \mid f_n(x) \ge c \cdot s(x)\}.$$

Then, each E_n is a preimage of $[0,\infty]$ by $f_n - c\cdot s$, so measurable, and $f_n \le f_{n+1}$ implies $E_n \subseteq E_{n+1}$ for each $n \in \mathbb{N}$. Moreover,

$X = \bigcup_{n=1}^{\infty} E_n$ because: given $x \in X$, $f(x) = 0$ implies $s(x) = 0$ so $x \in E_1$, and if $f(x) > 0$, then $c \cdot s(x) < s(x) \le f(x)$ and since $f_n(x) \to f(x)$, $x \in E_n$ for large enough $n \in \mathbb{N}$. Also, for each $n \in \mathbb{N}$,

$$c \cdot \int_{E_n} s \, d\mu \le \int_{E_n} f_n \, d\mu \le \int_X f_n \, d\mu$$

And, since the integral of simple function is countably additive, so

$$\int_X s \, d\mu = \int_{\bigcup_{n=1}^{\infty} E_n} s \, d\mu = \lim_{n\to\infty} \int_{E_n} s \, d\mu, \text{ so } c \cdot \int_X s \, d\mu \le \alpha$$

Since $c \in (0,1)$ was arbitrary, $\int_X s \, d\mu \le \alpha$.

And, the s is also arbitrary simple function satisfying $0 \le s \le f$,
Consequently $\int_X f \, d\mu \le \alpha$, so proved. ∎

Fatou's Lemma.

If $f_n : X \to [0, \infty]$ is measurable for $n \in \mathbb{N}$, then

$$\int_X \left(\liminf_{n\to\infty} f_n \right) d\mu \le \liminf_{n\to\infty} \int_X f_n \, d\mu$$

Proof. For each $k \in \mathbb{N}$, define $g_k(x) = \inf_{i \ge k} f_i(x)$.

Directly, $g_k \le f_k$ for each $k \in \mathbb{N}$, so that $\int_X g_k \, d\mu \le \int_X f_k \, d\mu$.

And, $0 \le g_1 \le g_2 \le \cdots$ by definition and g_k is measurable by theorem, the monotone convergence theorem shows that

$$\lim_{n\to\infty} \int_X g_n \, d\mu \overset{M.C.T}{=} \int_X \lim_{n\to\infty} g_n \, d\mu \overset{def}{=} \int_X \left(\liminf_{n\to\infty} f_n \right) d\mu \quad \cdots (*)$$

Meanwhile, by the monotonicity of g_k, for each $n \in \mathbb{N}$,

$$\inf_{k \ge n} \int_X g_k \, d\mu = \int_X g_n \, d\mu, \text{ so that}$$

$$\lim_{n\to\infty} \int_X g_n \, d\mu = \liminf_{n\to\infty} \int_X g_n \, d\mu \le \liminf_{n\to\infty} \int_X f_n \, d\mu$$

Now, the equality $(*)$ shows the result. ∎

Thm. Let $f_n : X \to [0, \infty]$ $(n \in \mathbb{N})$ be measurable functions with

$$f(x) = \sum_{n=1}^{\infty} f_n(x) \quad \text{exists for all } x \in X.$$

Then, $\int_X f \, d\mu = \sum_{n=1}^{\infty} \int_X f_n \, d\mu$.

Proof. First, there are monotone increasing sequence of simple functions $\{s_n'\}$ and $\{s_n''\}$ such that $s_n' \to f_1$ and $s_n'' \to f_2$.

Thm. Suppose $f : X \to [0, \infty]$ is measurable and $\varphi(E) := \int_E f \, d\mu$ $(E \in \mathcal{M})$.

Then, φ is a measure on $\mathcal{M}$, and

for any measurable $g : X \to [0, \infty]$, $\int_X g \, d\varphi = \int_X gf \, d\mu$.

Proof. Let $E_n \in \mathcal{M}$ $(n \in \mathbb{N})$ be pairwise disjoint measurable sets.

observe $\chi_E \cdot f = \sum_{n=1}^{\infty} \chi_{E_n} \cdot f$ where $E = \bigcup_{n=1}^{\infty} E_n$